

VOCATIONAL TRAINING REPORT

On

Chhattisgarh State Power Distribution
Company Limited (CSPDCL)



Submitted by:

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Branch: Electrical and Electronics Engineering (EEE), 5th Semester

Institute: Government Engineering College, Raipur

Training Location: CSPDCL, Nawapara Rajim

Training Duration: 01-07-2025 to 31-07-2025

Declaration

I, Aman Mehta, Roll No. 301602523002, student of Electrical and Electronics Engineering (EEE), 5th Semester at Government Engineering College, Raipur, hereby declare that this Vocational Training Report on CSPDCL, Nawapara Rajim Substation is my own original work.

This report has been prepared under the guidance of. _____, (AE),
CSPDCL, Nawapara Rajim Substation

I affirm that all the information included in this report is true to the best of my knowledge.

Signature:

Name: Aman Mehta

Roll No.: 301602523002.

ACKNOWLEDGEMENT

I am grateful to CSPDCL, 33/11 kV Sub-Station, AE (D), Nawapara-Rajim, C.G., for providing me the opportunity to undergo vocational training. This training helped me gain practical exposure to sub-station operations, equipment, and the power distribution system.

Sub-Station Equipment and Supply Arrangements:

Incomers: The sub-station has 2 incomers at 33 kV. The first incomer is continuous, while the second incomer functions as a feedback incomer.

Feeders: The sub-station has a total of 12 feeders—4 at 33 kV and 8 at 11 kV.

33 kV feeders: 4 feeders supplying the distribution system.

11 kV feeders:

1. 1st feeder
2. 2nd feeder
3. 3rd feeder
4. 4th feeder
5. 5th feeder
6. 6th feeder
7. 7th feeder
8. 8th feeder

Power Transformers: The sub-station has two power transformers:

1. 5 MVA rating

Metering Equipment (ME): Works in combination with Current Transformers (CT) and Potential Transformers (PT) to measure incoming and outgoing currents and voltages.

CT (Current Transformer): 0.5 rating, used for measuring supply current.

PT (Potential Transformer): 1.0 rating, used for measuring supply voltage.

Circuit Breakers: Mostly vacuum circuit breakers are used for protection.

Insulators: Post insulators and disc insulators are primarily used.

Conductors: Bare ACSR conductors are used for the distribution system.

Earthing: Pipe-type earthing is implemented to protect against earth faults.

This practical exposure allowed me to understand the design, operation, and maintenance of a 33/11 kV sub-station and its critical role in ensuring reliable power supply.

Signature: _____

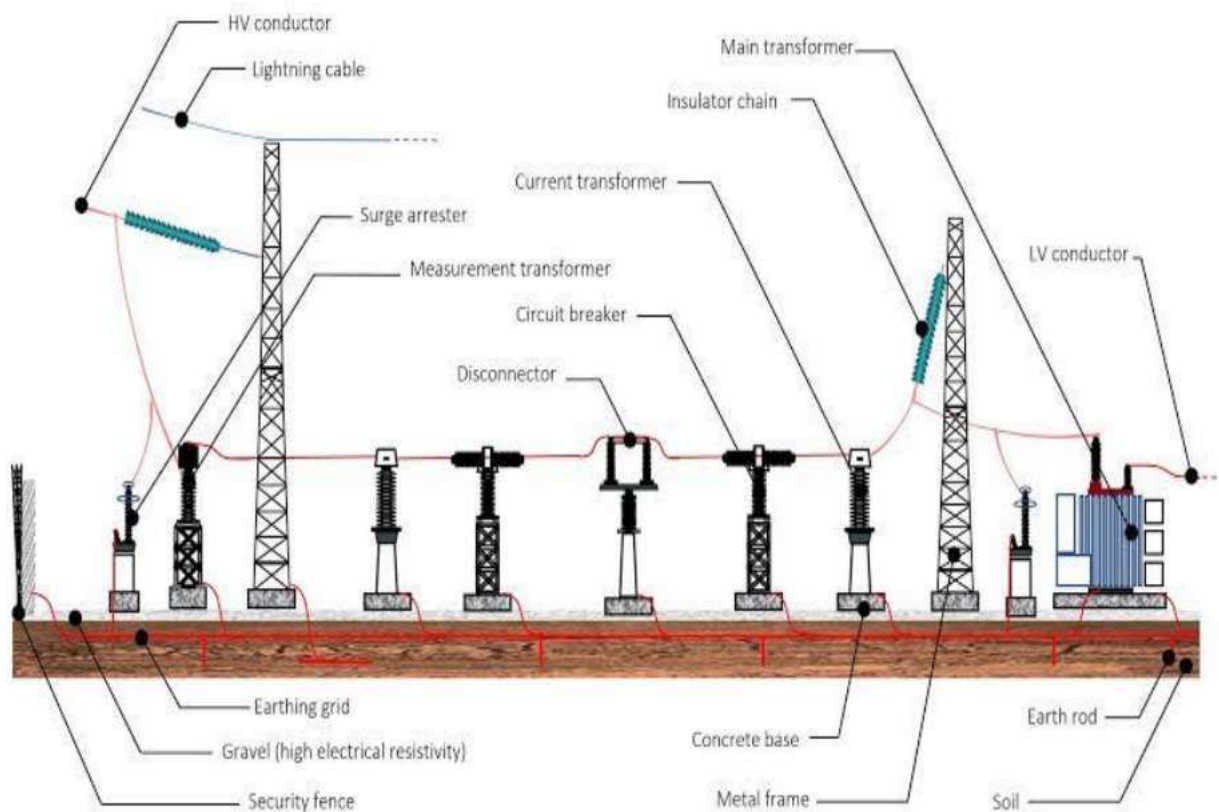
Name: Aman Mehta

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INTRODUCTION OF SUB-STATION

A substation is a high-voltage electric system facility. It is used to switch generators, equipment, and circuits or lines in and out of a system. It also is used to change AC voltages from one level to another, and/or change alternating current to direct current or direct current to alternating current.



Some substations are small with little more than a transformer and associated switches. Others are exceptionally large with several transformers and dozens of switches and other equipment.

SUBSTATION CHARACTERISTICS:

- Each circuit is protected by its own circuit breaker and hence plant outage does not necessarily result in loss of supply.
- A fault on the feeder or transformer circuit breaker causes loss of the transformer and feeder circuit, one of which may be restored after isolating the faulty circuit breaker.
- A fault on the bus section circuit breaker causes complete shutdown of the substation. All circuits may be restored after isolating the faulty circuit breaker.
- Maintenance of a feeder or transformer circuit breaker involves loss of the circuit.
- Introduction of bypass isolators between bus bar and circuit isolator allows circuit breaker maintenance facilities without loss of that circuit.

WORKING OF SUB-STATION

The present-day electrical power system is A.C.i.e. electrical power is generated, transmitted & distributed in the form of the alternating current. The electric power is produced at power plant stations which are located at favourable places generally quite away from the consumers.

It is delivered to the consumers through a large network of transmission & distribution. At many places in the power system, it may be desirable and necessary to change some.

characteristics e.g., voltage, ac to dc, frequency, power factor etc. of electric supply. This is accomplished by suitable apparatus called substation. For example, generation voltage (11 KV or 33 KV) at the power station is set up to high voltage (say 220 KV or 132 KV) for transmission of electric power.

The assembly of apparatus (e.g., transformer etc.) used for this purpose in the substation. Similarly, near the consumer's localities, the voltage may have to be stepped down to utilization level. This job is again accomplished by suitable apparatus called substation. The assembly of apparatus to change some characteristic of electric power supply is called substation.

The two most ways to classify substation are: -

➤ **According to the service requirement:**

1. Transformer substation
2. Switch substation
3. Power factor correction substation
4. Converting substation
5. Industrial substation

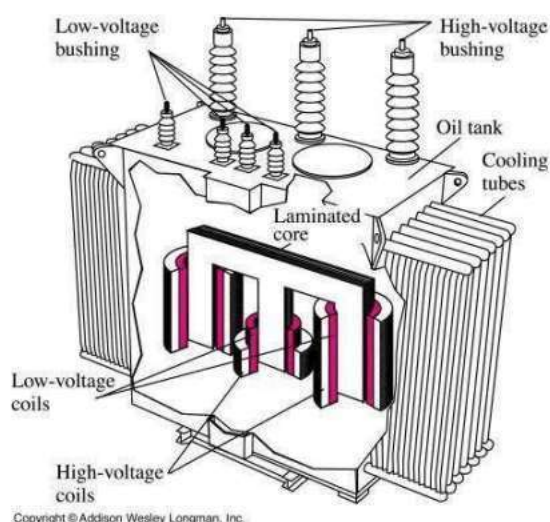
➤ **According to the constructional features:**

1. Indoor substation
2. Outdoor substation
3. Underground substation
4. Pole mounted substation.

EQUIPMENT OF SUBSTATION

- Power Transformer
- C.T. {Current Transformer}
- P.T. {Potential Transformer}
- Circuit Breaker
- Insulator
- Conductor
- Pannel Board

Power Transformer:-

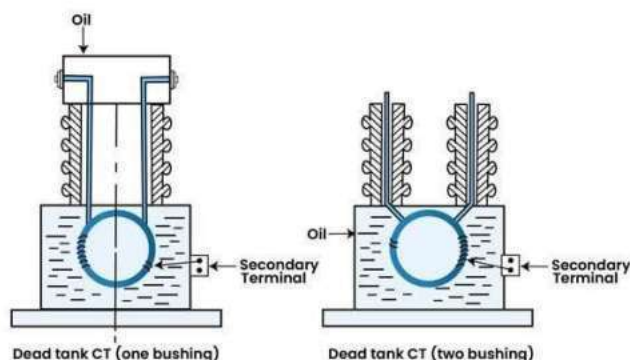


It is used for the transmission purpose at heavy load, high voltage greater than 33 KV & 100% efficiencies. It also having a big in size as compared to distribution transformer, it used in generating station and Transmission substation

at high insulation level. They can be of two types: Single Phase Transformers and Multi Phase Transformers. These are on load tap changing transformer and have star winding connection.

C.T. {Current Transformer}:-

A current transformer (CT) is used for measurement of alternating

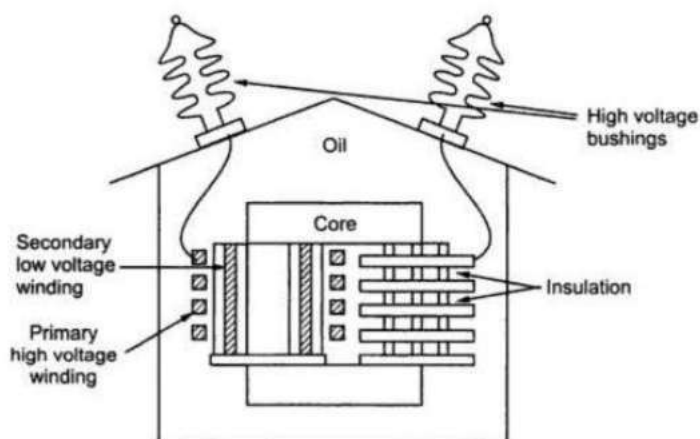


electric currents. When current in a circuit is too high to apply directly to measuring instruments, a current transformer produces a reduced current

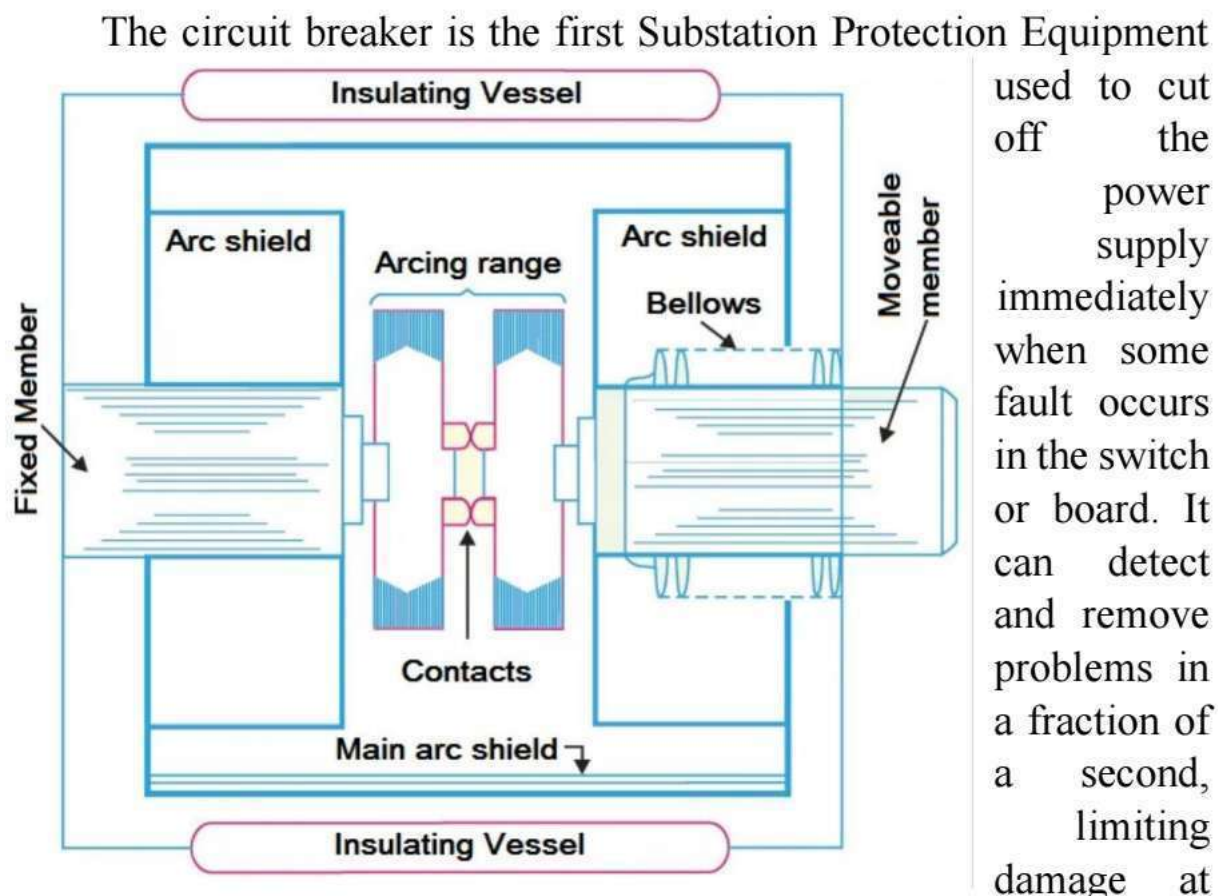
accurately proportional to the current in the circuit, which can be conveniently connected to measuring and recording instruments. A current transformer isolates the measuring instruments from what may be extremely high voltage in the monitored circuit.

P.T. {Potential Transformer}:-

Voltage transformers (VT) (also called potential transformers (PT)) are a parallel connected type of instrument transformer, used for metering and protection in high-voltage circuits or phasor phase shift isolation. They are designed to present negligible load to the supply being measured and to have an accurate voltage ratio to enable accurate metering. A potential transformer may have several secondary windings on the same core as a primary winding, for use in different metering or protection circuits.



Circuit Breaker:-



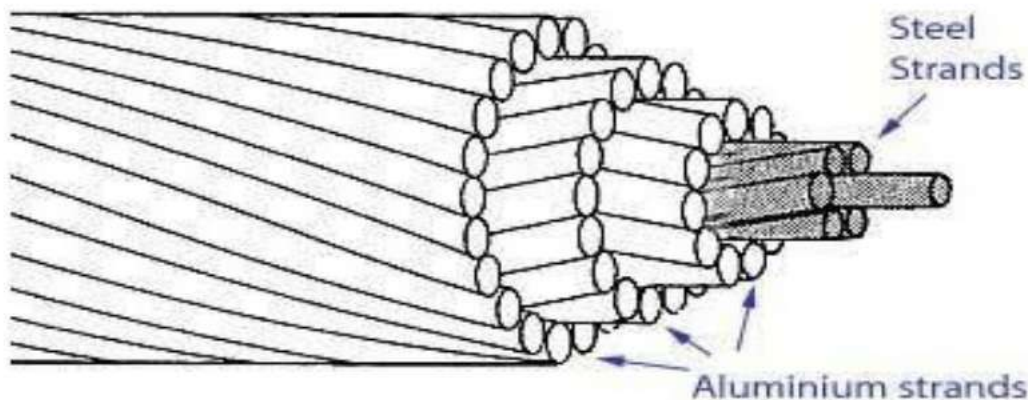
the point of failure. The circuit breakers interrupt extremely high fault currents, which can be ten times more than the regular operating currents.

TYEPS of Circuit Breaker: -

- Air blast circuit breaker
- Oil circuit breaker
- Sulphur hexafluoride circuit breaker (SF6) circuit breaker
- Vacuum circuit breaker
- MCCB (Molded Case Circuit Breaker)

CONDUCTOR:-

there are used the bare conductor ACSR is used to distribution system.

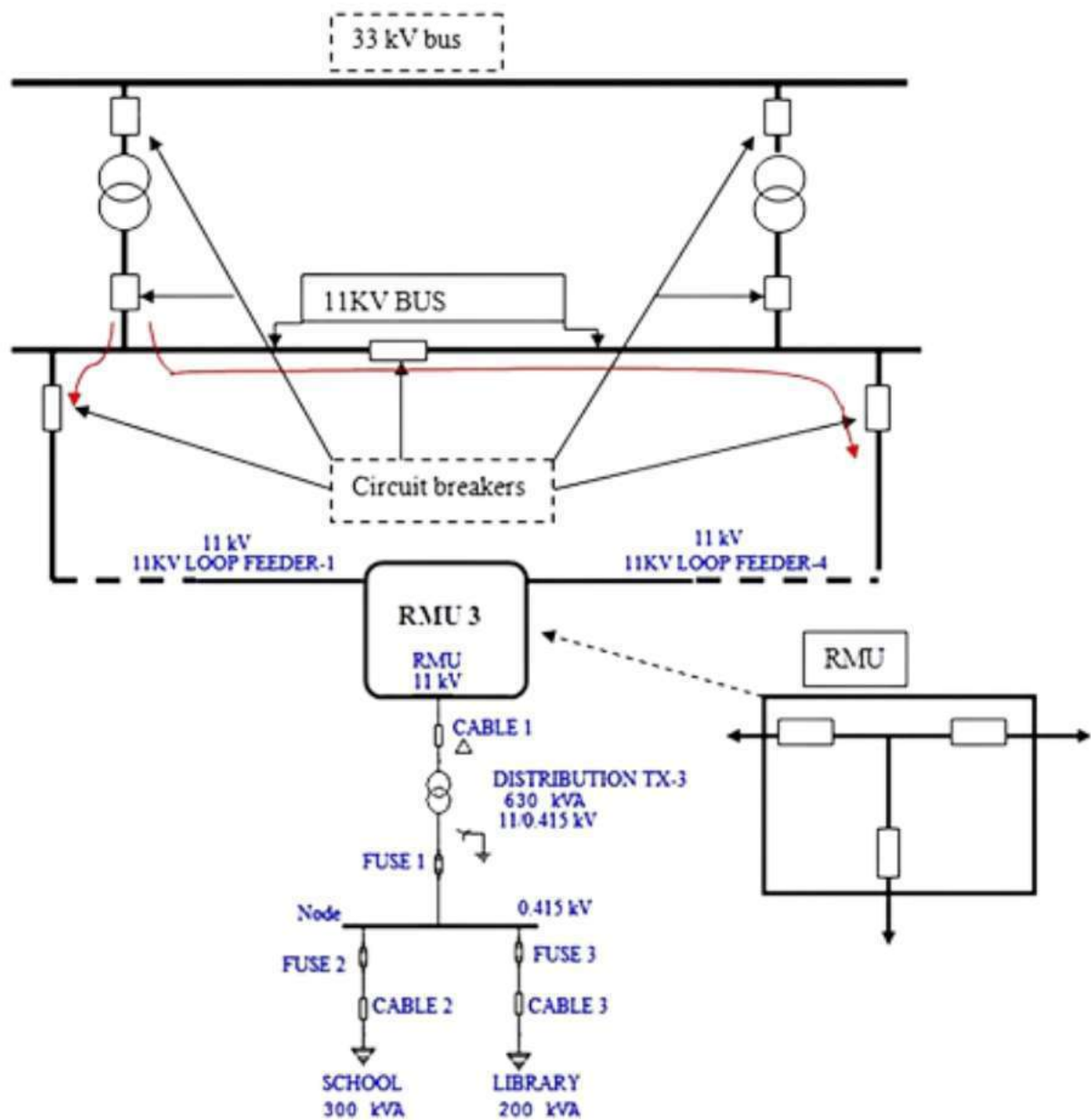


ACSR: - Aluminium conductor steel-reinforced cable (ACSR) is a type of high-capacity, high-strength stranded conductor typically used in overhead power lines. The outer strands are high-purity aluminium, chosen for its good conductivity, low weight, low cost, resistance to corrosion and decent mechanical stress resistance. The centre strand is steel for additional strength to help support the weight of the conductor. Steel is of higher strength than aluminium which allows for increased mechanical tension to be applied on the conductor. Steel also has lower elastic and inelastic deformation (permanent elongation) due to mechanical loading (e.g., wind and ice) as well as a lower coefficient of thermal expansion under current loading. These properties allow ACSR to sag significantly less than all-aluminium conductors. As per the International Electrotechnical Commission (IEC) and The CSA Group (formerly the Canadian Standards Association or CSA) naming convention, ACSR is designated.

TYPES OF BARE CONDUCTOR :-

- **AAC** (All Aluminium Conductor)
- **AAAC** (All Aluminium Alloy Conductor)
- **ACSR** (Aluminium Conductor Steel Reinforced)
- **ACAR** (Aluminium Conductor Alloy Reinforced)

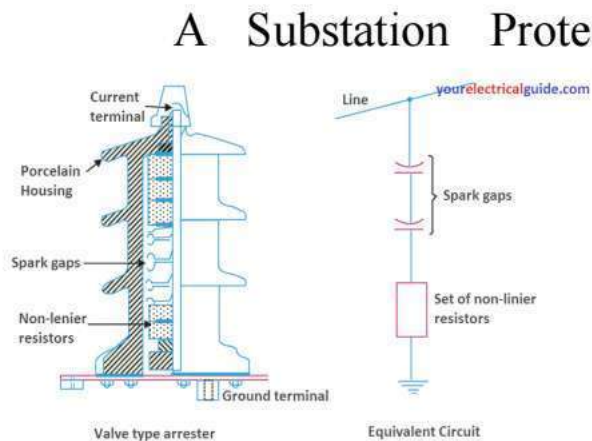
BLOCK DIAGRAM OF SUBSTATION



PROTECTION SYSTEM OF SUB-STATION

- Lightning arrester
- Earthing

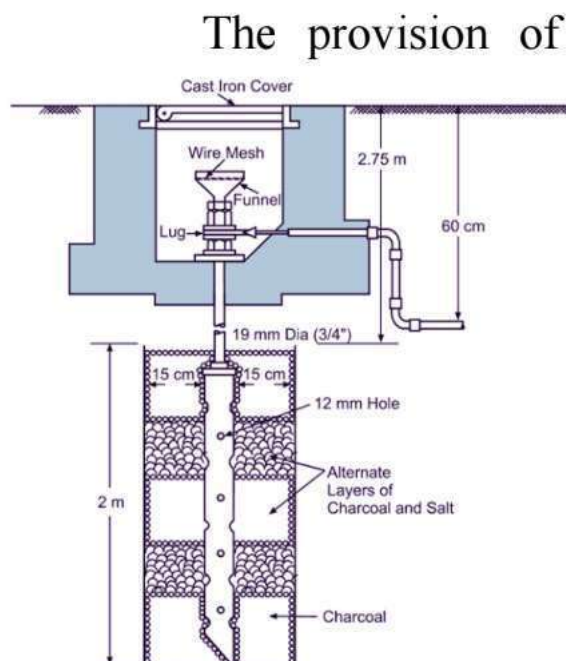
1. Lightning arrester:



A Substation Protection Equipment's lightning arrester is a critical safety device protecting valuable equipment and workers. During lightning, it arrests and discharges overvoltage to the ground. These are present between the line and the

ground near the system.

2. Earthing:



The provision of an effective, long-lasting, and dependable earthing system in a substation and switching station is critical for the safety of workers and electrical equipment. The voltage levels do not exceed acceptable limits. The earth connection is sturdy enough to dissipate the fault to the ground done to Substation Protection Equipment. Earthing

connects electrical objects to the earth's general mass and has very less resistance.

Conclusion

The vocational training at 33/11 kV Nawapara-Rajim sub-station has been a highly enriching and transformative experience. It gave me a clear and practical understanding of the distribution side of the power sector, which plays a vital role in supplying electricity to consumers reliably and efficiently. During my training, I was introduced to the working of essential components such as power transformers, circuit breakers, isolators, relays, lightning arresters, fuses, insulators, and bus bar. I learned how these devices are interconnected and coordinated to ensure safe and smooth operation of the sub-station.

Beyond theoretical knowledge, the practical exposure was extremely valuable. I got the opportunity to observe and understand fault detection methods, transformer maintenance practices, load management, cooling system monitoring, and emergency restoration techniques. These experiences bridged the gap between classroom concepts and real-world applications, enabling me to understand the importance of timely maintenance, safety procedures, and teamwork in the power industry.

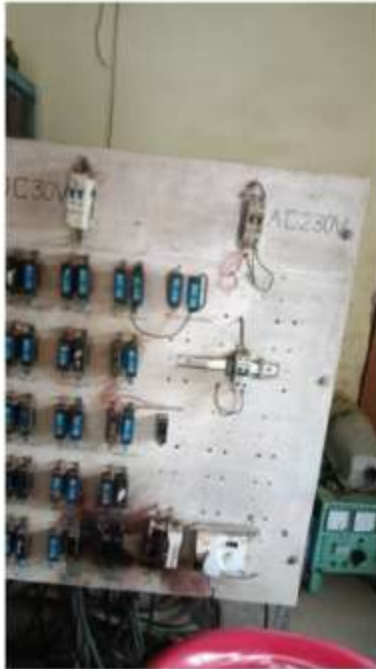
This training not only enhanced my technical knowledge but also helped me develop professional discipline, problem-solving ability, and confidence to handle real-time electrical systems. The exposure to actual working conditions and troubleshooting methods has given me a broader perspective of the responsibilities and challenges faced by engineers in the power sector.

Overall, the training at the 33/11 kV Nawapara Rajim sub-station has been an invaluable learning journey. It has strengthened my foundation as an electrical engineering student and will serve as a stepping stone for my future academic and professional career. The knowledge, skills, and insights gained during this period will undoubtedly guide me in becoming a competent and responsible electrical engineer.

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Thank You CSPDCL